

DATA MINING LAB

A Regional Correlation Study of Household Economics and School Enrollment in the Philippines

Report Documentation - Final

Submitted by Group 20:

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December 8, 2025

I. Introduction

This study will analyze the relationship between household economic conditions and school enrollment across Philippine regions. It will combine two public datasets sourced from Kaggle.com that contains a total of three tables: (1) a K–10 enrollment dataset containing regional enrollment counts by grade level and school year, (2) a Senior High School (SHS) enrollment dataset containing Grade 11 and Grade 12 enrollment counts by strand and region, and (3) a Filipino Family Income and Expenditure Survey (FFIES) dataset that provides information on household income, food and non-food expenditures, and demographic characteristics. By linking these datasets at the regional level, the project seeks to identify patterns linking household income and spending behavior with enrollment levels and SHS strand choice in both basic education and senior high school level. The purpose of this project is to apply reproducible data-mining and statistical techniques to produce interpretable insights that can inform education stakeholders about the potential influence of economic conditions of Filipino families on school participation per region.

1.1 Dataset Description

1. Filipino Family Income and Expenditure
 - a. **Source:** Filipino Family Income and Expenditure Dataset by Philippine Statistics Authority (PSA), made available through Kaggle (uploaded by Francis Paul Flores).
 - b. **Coverage:** Nationwide household-level data from the Philippines, covering all administrative regions. The dataset represents individual households and includes information on income, detailed expenditure categories, demographic characteristics, housing conditions, and ownership of household assets.
 - c. **Type of Data** (Data Dictionary)

Column Name	Data Type	Description	Unit
Total Household Income	Numeric	Total annual household income from all sources	Philippine Peso (PHP)
Region	Text	Geographic region where the household is located	–
Total Food Expenditure	Numeric	Total household spending on food	PHP
Main Source of Income	Text	Primary source of household income	–



		(e.g., wages, business)	
Agricultural Household Indicator	Numeric	Indicator whether household is engaged in agriculture (1 = Yes, 0 = No)	Binary
Bread and Cereals Expenditure	Numeric	Household expenditure on bread and cereals	PHP
Total Rice Expenditure	Numeric	Household expenditure on rice	PHP
Meat Expenditure	Numeric	Household expenditure on meat products	PHP
Total Fish and Marine Products Expenditure	Numeric	Household expenditure on fish and marine products	PHP
Fruit Expenditure	Numeric	Household expenditure on fruits	PHP
Vegetables Expenditure	Numeric	Household expenditure on vegetables	PHP
Restaurant and Hotels Expenditure	Numeric	Spending on restaurants and hotel food services	PHP
Alcoholic Beverages Expenditure	Numeric	Household spending on alcoholic beverages	PHP
Tobacco Expenditure	Numeric	Household spending on tobacco products	PHP



Clothing, Footwear and Other Wear Expenditure	Numeric	Household spending on clothing and footwear	PHP
Housing and Water Expenditure	Numeric	Household spending on housing and water services	PHP
Imputed House Rental Value	Numeric	Estimated rental value of owner-occupied housing	PHP
Medical Care Expenditure	Numeric	Household spending on medical and health services	PHP
Transportation Expenditure	Numeric	Household spending on transportation	PHP
Communication Expenditure	Numeric	Household spending on communication services	PHP
Education Expenditure	Numeric	Household spending on education	PHP
Miscellaneous Goods and Services Expenditure	Numeric	Household spending on miscellaneous goods and services	PHP
Special Occasions Expenditure	Numeric	Spending related to special events and occasions	PHP
Crop Farming and Gardening Expenses	Numeric	Expenses related to crop farming and gardening	PHP
Total Income from Entrepreneurial Activities	Numeric	Income generated from household entrepreneurial activities	PHP



Household Head Sex	Text	Sex of the household head	–
Household Head Age	Numeric	Age of the household head	Years
Household Head Marital Status	Text	Marital status of the household head	–
Household Head Highest Grade Completed	Text	Highest educational attainment of the household head	–
Household Head Job or Business Indicator	Text	Employment or business participation of household head	–
Household Head Occupation	Text	Occupation of the household head	–
Household Head Class of Worker	Text	Employment class of the household head	–
Type of Household	Text	Household classification (e.g., nuclear, extended)	–
Total Number of Family Members	Numeric	Total number of individuals living in the household	Count
Members with Age Less Than 5 Years Old	Numeric	Number of household members aged below 5	Count
Members with Age 5–17 Years Old	Numeric	Number of household members aged 5 to 17	Count



Total Number of Family Members Employed	Numeric	Number of employed household members	Count
Type of Building/House	Text	Structural type of the dwelling	–
Type of Roof	Text	Material used for the house roof	–
Type of Walls	Text	Material used for house walls	–
House Floor Area	Numeric	Total floor area of the house	Square meters
House Age	Numeric	Age of the house	Years
Number of Bedrooms	Numeric	Total number of bedrooms	Count
Tenure Status	Text	Ownership or tenure status of the dwelling	–
Toilet Facilities	Text	Type of toilet facility used by the household	–
Electricity	Numeric	Indicator of access to electricity (1 = Yes, 0 = No)	Binary
Main Source of Water Supply	Text	Primary source of household water	–
Number of Television	Numeric	Number of television units owned	Count
Number of CD/VCD/DVD	Numeric	Number of CD/VCD/DVD players owned	Count



Number of Component/Stereo Set	Numeric	Number of component or stereo sets owned	Count
Number of Refrigerator/Freezer	Numeric	Number of refrigerators or freezers owned	Count
Number of Washing Machine	Numeric	Number of washing machines owned	Count
Number of Air Conditioner	Numeric	Number of air conditioning units owned	Count
Number of Car, Jeep, Van	Numeric	Number of cars, jeeps, or vans owned	Count
Number of Landline/Wireless Telephones	Numeric	Number of landline or wireless telephones owned	Count
Number of Cellular Phone	Numeric	Number of cellular phones owned	Count
Number of Personal Computer	Numeric	Number of personal computers owned	Count
Number of Stove with Oven/Gas Range	Numeric	Number of stoves with oven or gas range owned	Count
Number of Motorized Banca	Numeric	Number of motorized bancas owned	Count
Number of Motorcycle/Tricycle	Numeric	Number of motorcycles or tricycles owned	Count



2. Filipino Family Income and Expenditure

- a. **Source:** Philippines School Enrollment Data, compiled from official Department of Education (DepEd) enrollment statistics and made available through Kaggle (uploaded by Paul Ryan Lacsamana).
- b. **Coverage:** Nationwide school enrollment data covering all administrative regions of the Philippines across multiple academic years. The dataset contains enrollment counts by region, academic year, and education sector (public, private, and SUCs/LUCs), with detailed breakdowns by grade level from Kindergarten to Grade 12, including Senior
- c. **Type of Data** (Data Dictionary)

Column Name	Data Type	Description	Unit
item_no	Numeric / Integer	Unique record identifier	–
sector	Text	Type of educational institution (public, private, SUC-LUC)	–
Academic_Year	Text	Academic year of record (e.g. SY 2010-2011)	–
AY_Start	Numeric / Integer	Start year of academic period	Year
AY_End	Numeric / Integer	End year of academic period	Year
Region	Text	Administrative region of the Philippines	–
Kindergarten	Numeric / Integer	Number of enrolled students in Kindergarten	Count
Grade 1	Numeric / Integer	Number of enrolled students in Grade 1	Count



Grade 2	Numeric / Integer	Number of enrolled students in Grade 2	Count
Grade 3	Numeric / Integer	Number of enrolled students in Grade 3	Count
Grade 4	Numeric / Integer	Number of enrolled students in Grade 4	Count
Grade 5	Numeric / Integer	Number of enrolled students in Grade 5	Count
Grade 6	Numeric / Integer	Number of enrolled students in Grade 6	Count
Grade 7	Numeric / Integer	Number of enrolled students in Grade 7	Count
Grade 8	Numeric / Integer	Number of enrolled students in Grade 8	Count
Grade 9	Numeric / Integer	Number of enrolled students in Grade 9	Count
Grade 10	Numeric / Integer	Number of enrolled students in Grade 10	Count
Grade 11_ABM	Numeric / Integer	Number of enrolled students in Grade 11 — ABM strand	Count
Grade 11_HUMSS	Numeric / Integer	Number of enrolled students in Grade 11 — HUMSS strand	Count
Grade 11_STEM	Numeric / Integer	Number of enrolled students in Grade 11 — STEM strand	Count
Grade 11_GAS	Numeric / Integer	Number of enrolled students in Grade 11 — GAS strand	Count



Grade 11_MARITIME	Numeric / Integer	Number of enrolled students in Grade 11 — MARITIME strand	Count
Grade 11_TVL	Numeric / Integer	Number of enrolled students in Grade 11 — TVL strand	Count
Grade 11_SPORTS	Numeric / Integer	Number of enrolled students in Grade 11 — SPORTS strand	Count
Grade 11_A&D	Numeric / Integer	Number of enrolled students in Grade 11 — Arts & Design strand	Count
Grade 12_ABM	Numeric / Integer	Number of enrolled students in Grade 12 — ABM strand	Count
Grade 12_HUMSS	Numeric / Integer	Number of enrolled students in Grade 12 — HUMSS strand	Count
Grade 12_STEM	Numeric / Integer	Number of enrolled students in Grade 12 — STEM strand	Count
Grade 12_GAS	Numeric / Integer	Number of enrolled students in Grade 12 — GAS strand	Count
Grade 12_MARITIME	Numeric / Integer	Number of enrolled students in Grade 12 — MARITIME strand	Count

Grade 12_TVL	Numeric / Integer	Number of enrolled students in Grade 12 — TVL strand	Count
Grade 12_SPORTS	Numeric / Integer	Number of enrolled students in Grade 12 — SPORTS strand	Count
Grade 12_A&D	Numeric / Integer	Number of enrolled students in Grade 12 — Arts & Design strand	Count

1.2 Objectives of the Study

General Objective

To analyze how regional household socioeconomic characteristics influence basic and Senior High School (SHS) enrollment patterns across the Philippines by integrating income–expenditure data with K–10 and SHS enrollment statistics. The study aims to uncover regional differences, identify which socioeconomic factors most strongly relate to SHS track distribution, and derive insights using correlation analysis, clustering, association patterns, and decision-tree classification to support educational planning, policy formulation, and resource allocation.

Specific Research Questions

Specifically, this study aims to answer the following research questions:

1. How are K–10 and SHS enrollments distributed across regions, based on the available aggregated enrollment data?
2. Which socioeconomic factors most strongly influence SHS strand selection?
3. What correlations exist between household socioeconomic variables and school enrollment indicators, such as K–10 totals, SHS totals, and SHS strand counts?
4. Which SHS strands do students from regions with higher household income or higher education expenditure more likely to pursue?
5. Are there observable associations between income levels and the type of academic sector students enroll in?
6. What insights can be derived from the combined economic–education data that may help policymakers understand regional disparities in schooling and resource needs?

1.3 Planned Techniques and Tools Used

Tools Used

- a. **Microsoft Power BI** - Primary platform for dashboard development and interactive data visualization. Power BI was used to present regional comparisons of household income, expenditure patterns, and school enrollment indicators through charts, maps, and KPI visuals.
- b. **Canva** - Used as the primary dashboard design and presentation tool for creating professional, visually appealing, and easily understandable reports.
- c. **Altair AI Studio (RapidMiner)** - Served as the primary data mining and preprocessing platform. RapidMiner was used for data cleaning, transformation, normalization, aggregation, attribute generation, and dataset integration following the CRISP-DM methodology.
- d. **Microsoft Excel** - Utilized for initial data inspection, validation, and basic cleaning. Excel was used to quickly review raw CSV files, verify column consistency, inspect missing values, and perform preliminary sanity checks before importing the data into RapidMiner.
- e. **Kaggle** - Primary source for dataset acquisition. The Family Income and Expenditure Survey dataset and the Philippine School Enrollment datasets were sourced from Kaggle, ensuring publicly available and well-documented data for reproducibility.

Data Mining Methods and Techniques

1. Correlation Matrix
 - A correlation matrix was constructed to examine pairwise relationships between household socio-economic variables (income, food and non-food expenditures, education spending, family size) and Senior High School (SHS) track enrollments.
 - This method provides a foundational understanding of which socio-economic factors are most strongly associated with SHS strand selection and overall enrollment patterns.
2. k-Means Clustering
 - k-Means clustering was used to group regions based on similarities in household socio-economic characteristics and SHS track enrollments.
 - This unsupervised learning technique identifies natural clusters in the data without predefined labels, highlighting regions with similar household and student participation patterns.
3. Decision Tree Classification
 - Decision tree classification was employed to determine which SHS strand is most likely chosen based on household socio-economic attributes.
 - Key predictors include food expenditure share, education spending, household income, household head age, and school sector.
 - This method identifies the most influential factors driving enrollment in specific SHS tracks.

II. Data Processing

2.1 Overview of Raw Data Issues

Both datasets used in this project had raw data issues before they could be used for analysis:

1. Family Income and Expenditure Survey

- 1.1. Some numeric fields such as income, expenditures, and household counts had missing values. These needed to be replaced or imputed.
- 1.2. Duplicates existed in household records and needed to be removed.
- 1.3. Some numeric fields were recorded as text and required correction.
- 1.4. Certain attributes were irrelevant for the analysis (e.g., household appliances, housing materials).

2. Philippine School Enrollment Datasets (K–10, SHS, SHS Strand per Region)

- 2.1. Missing enrollment values for some grades and SHS strands.
- 2.2. Duplicate records for some regions and academic years.
- 2.3. Inconsistent region names across datasets.
- 2.4. Irrelevant columns such as identifiers or extended school information.

2.2 Data Cleaning Steps

1. Imported all datasets into the data mining platform (Altair AI Studio / RapidMiner). Below is a list of processes present in the data mining process and how it was used.
2. Handling Missing Values:
 - Replaced missing numeric values (income, expenditures, enrollment counts) with 0 or mean depending on relevance.
 - Ensured no blank region fields to avoid join errors.
3. Removing Duplicates:
 - Removed repeated household rows in the Family Income and Expenditure dataset.
 - Verified that school enrollment datasets contained no duplicates.
4. Correcting Data Types:
 - Convert numeric-looking strings to real values.
 - Standardized categorical attributes (region, sector) into polynomial format.
5. Filtering by Year:
 - Filtered all school datasets to include only records with AY_Start = 2017.

6. Region Standardization:

- Normalized region names across datasets (e.g., “Region I – Ilocos Region” to “Ilocos Region”).

7. Removing Irrelevant Attributes:

- Dropped identifiers and columns unrelated to the analysis (household appliances, housing materials).

2.3 Data Transformation

1. Normalization:

- Applied Z-Transformation to income and expenditure variables to reduce skewness and allow fair comparison with enrollment values.

2. Aggregation of Enrollment Data:

- Calculated total K–10 enrollment per region.
- Calculated total Grade 11 + Grade 12 enrollment per region.
- Summed total enrollment per SHS strand (ABM, HUMSS, STEM, GAS, TVL, SPORTS, A&D).

3. Derived Attributes:

- $\text{Food_Share} = \text{Total Food Expenditure} / \text{Total Household Expenditure}$
- $\text{Education_Share} = \text{Education Expenditure} / \text{Total Household Expenditure}$
- $\text{NonFood_Expenditure} = \text{Total Household Expenditure} - \text{Total Food Expenditure}$
- $\text{Total_SHS} = \text{Grade11_Total} + \text{Grade12_Total}$

4. Merging Datasets:

- Joined the Family Income and Expenditure dataset with enrollment aggregates using the Region field.

2.4 Final Dataset Description

After completing all cleaning, filtering, aggregation, and transformation procedures, the final integrated dataset consists of regional-level socio-economic indicators combined with K–10 and Senior High School (SHS) enrollment totals for Academic Year 2017–2018. The dataset contains averaged household income and expenditure variables from the Family Income and Expenditure Survey (FIES), merged with total enrollment counts from the DepEd School Enrollment dataset, including strand-level SHS distributions. All records correspond to 2017 only, and all socio-economic variables were aggregated by Region to ensure uniform granularity across datasets.

Number of rows: 51

Number of columns: 52



Column Name	Data Type
Region	Text
average(Clothing, Footwear and Other Wear Expenditure)	Numeric
average(Communication Expenditure)	Numeric
average(Education Expenditure)	Numeric
average(Electricity)	Numeric
average(Household Head Age)	Numeric
average(Housing and water Expenditure)	Numeric
average(Medical Care Expenditure)	Numeric
average(Members with age 5 - 17 years old)	Numeric
average(Members with age less than 5 year old)	Numeric
average(Miscellaneous Goods and Services Expenditure)	Numeric
average(Number of Cellular phone)	Numeric
average(Number of Motorcycle/Tricycle)	Numeric
average(Number of Motorized Banca)	Numeric
average(Number of Personal Computer)	Numeric
average(Total Food Expenditure)	Numeric
average(Total Household Income)	Numeric
average(Total Number of Family members)	Numeric
average(Total number of family members employed)	Numeric



average(Transportation Expenditure)	Numeric
item_no	Numeric
sector	Text
Academic_Year	Text
AY_Start	Numeric
AY_End	Numeric
Kindergarten	Numeric
Grade 1	Numeric
Grade 2	Numeric
Grade 3	Numeric
Grade 4	Numeric
Grade 5	Numeric
Grade 6	Numeric
Grade 7	Numeric
Grade 8	Numeric
Grade 9	Numeric
Grade 10	Numeric
Grade 11 ABM	Numeric
Grade 11 HUMSS	Numeric
Grade 11 STEM	Numeric
Grade 11 GAS	Numeric
Grade 11 MARITIME	Numeric
Grade 11 TVL	Numeric



Grade 11 SPORTS	Numeric
Grade 11 A&D	Numeric
Grade 12 ABM	Numeric
Grade 12 HUMSS	Numeric
Grade 12 STEM	Numeric
Grade 12 GAS	Numeric
Grade 12 MARITIME	Numeric
Grade 12 TVL	Numeric
Grade 12 SPORTS	Numeric
Grade 12 A&D	Numeric

III. Data Analysis Results

3.1 Overview of Techniques Applied

1. Correlation Matrix

Using correlation matrix, we will be able to identify and quantify the relationships between household socioeconomic variables and the distribution of students across various Senior High School tracks. Since the dataset includes indicators such as total household income, food and non-food expenditure, education spending, and the total number of family members, alongside counts of students enrolled in different SHS strands, the correlation matrix helps determine which socioeconomic factors are most strongly associated with track selection.

2. k-Means Clustering

Using k-Means clustering, we group regions based on similarities in their socioeconomic characteristics and Senior High School (SHS) track enrollments. This method helps identify natural groupings in the dataset without any predefined labels, allowing us to see which regions share similar household patterns or SHS participation behavior.

3. Decision Tree Classification

The decision tree model is used to determine which Senior High School (SHS) strand is most likely to be chosen based on household socioeconomic characteristics. By examining how variables such as food share, education spending, household income, age structure, and school sector relate to track selection, the model identifies the most important predictors that explain why a region or group tends to choose GAS, TVL, STEM, or ABM

3.2 Findings from the Data Mining Techniques Used

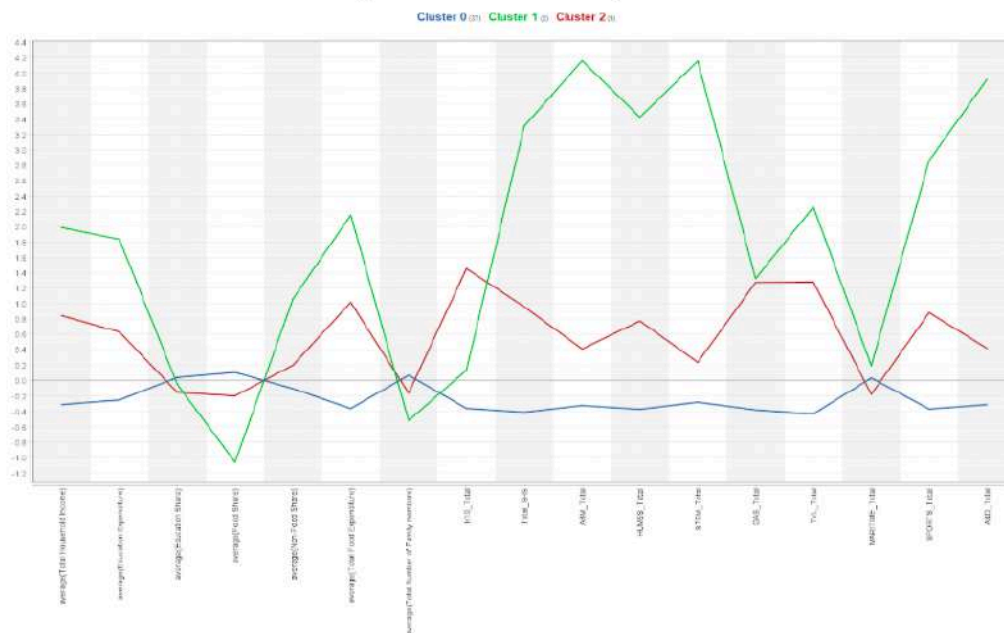
1. Correlation Matrix

Attributes	average...	average...	average...	average...	average...	average...	K10_To...	Total_S...	ABM_T...	HUMSS...	STEM...	GAS_T...	TVL_Ta...	MARITI...	SPORT...	A&D_T...
average(Total Household Income)	1	0.981	-0.712	0.712	0.935	-0.488	0.245	0.433	0.540	0.463	0.506	0.131	0.338	0.017	0.542	0.553
average(Education Expenditure)	0.981	1	-0.022	0.622	0.810	-0.370	0.255	0.361	0.404	0.399	0.431	0.090	0.280	-0.036	0.463	0.473
average(Food Share)	-0.712	-0.622	1	-1.000	-0.460	0.670	-0.053	-0.161	-0.267	-0.219	-0.245	0.067	-0.077	0.102	-0.252	-0.256
average(Non Food Share)	0.712	0.622	-1.000	1	0.460	-0.670	0.053	0.161	0.267	0.219	0.245	-0.067	0.077	-0.102	0.252	0.256
average(Total Food Expenditure)	0.935	0.810	-0.460	0.460	1	-0.315	0.310	0.506	0.502	0.513	0.553	0.208	0.420	0.043	0.563	0.581
average(Total Number of Family members)	-0.488	-0.370	0.670	-0.670	-0.315	1	-0.081	-0.170	-0.165	-0.198	-0.162	-0.082	-0.159	-0.079	-0.112	-0.129
K10_Total	0.245	0.255	-0.053	0.053	0.310	-0.081	1	0.585	0.235	0.516	0.118	0.776	0.791	-0.128	0.605	0.266
Total_SHS	0.433	0.361	-0.161	0.161	0.506	-0.170	0.585	1	0.887	0.959	0.839	0.816	0.941	0.172	0.761	0.812
ABM_Total	0.540	0.404	-0.267	0.267	0.592	-0.165	0.235	0.887	1	0.898	0.982	0.476	0.691	0.220	0.674	0.871
HUMSS_Total	0.463	0.399	-0.219	0.219	0.513	-0.198	0.516	0.959	0.898	1	0.947	0.695	0.977	0.142	0.881	0.759
STEM_Total	0.506	0.431	-0.245	0.245	0.553	-0.162	0.118	0.839	0.982	0.947	1	0.412	0.610	0.325	0.584	0.832
GAS_Total	0.131	0.090	0.067	-0.067	0.208	-0.082	0.776	0.816	0.476	0.695	0.412	1	0.927	0.993	0.884	0.827
TVL_Total	0.338	0.280	-0.077	0.077	0.420	-0.159	0.791	0.941	0.691	0.877	0.610	0.927	1	0.949	0.741	0.662
MARITIME_Total	0.017	-0.036	0.102	-0.102	0.043	-0.079	-0.128	0.172	0.220	0.142	0.325	0.090	0.949	1	-0.091	0.177
SPORTS_Total	0.542	0.463	-0.252	0.252	0.563	-0.112	0.605	0.761	0.674	0.681	0.584	0.884	0.741	-0.091	1	0.838
A&D_Total	0.553	0.473	-0.256	0.256	0.581	-0.129	0.266	0.812	0.871	0.759	0.832	0.527	0.662	0.177	0.838	1

Looking at the results of our correlation matrix, we can see that:

- Higher household income is moderately to strongly correlated with higher enrollment in several SHS tracks (ABM, STEM, SPORTS, A&D).
- Education expenditure shows similar positive correlations, suggesting that families who spend more on education tend to have students in specialized tracks.
- Food share has negative correlations with income-related variables, indicating that lower-income households allocate a larger portion of their budget to food.
- Tracks such as ABM, HUMSS, STEM, GAS, SPORTS, A&D are strongly correlated with each other, meaning regions with high participation in one strand often have high participation in others.
- K10 and Total SHS counts are positively correlated with almost all track totals, which is expected because they are aggregates.

2. k-Means Clustering



Based on the centroid values shown in the output:

Cluster 0 (25 records)

- Smaller family size
- Lower food share
- Lower MARITIME enrollment
- Interpretation: These regions generally have smaller households and less reliance on food expenditure, reflected in lower participation in the Maritime strand.

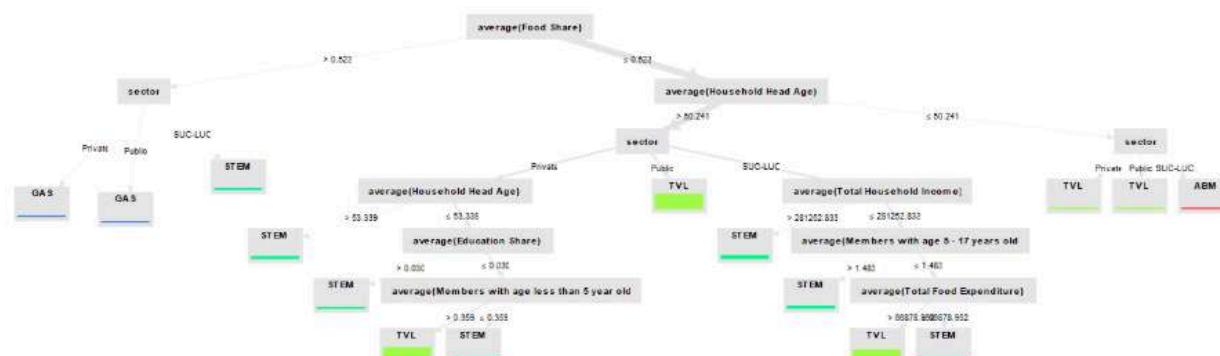
Cluster 1 (21 records)

- Larger family size
- Higher food share
- Much lower ABM enrollment
- Interpretation: These regions represent larger, likely lower-income households (high food share). They also show significantly low ABM participation.

Cluster 2 (2 records)

- Extremely high A&D, ABM, and STEM enrollments
- Interpretation: This small cluster consists of highly specialized or urbanized regions where academic and creative SHS tracks are unusually popular.

3. Decision Tree



The decision tree shows that Food Share is the strongest predictor of strand choice. Regions with high food share are often an indicator of lower income, and are more likely to select GAS, while regions with lower food share show more diverse outcomes.

Among low-food share regions, Household Head Age becomes the next major factor. Areas with older household heads tend to choose STEM or ABM, while those with younger household heads, especially in public-school areas, frequently choose TVL.

School Sector also plays a significant role:

- Public schools lean toward TVL,
- Private and SUC-LUC schools favor STEM or ABM,
- High-food share in private regions still tends toward GAS.

Other socioeconomic variables such as income, education spending, and household age composition help refine the final classification. Higher income and higher education spending typically push regions toward STEM, while lower income tends to direct them toward TVL.

Overall, the tree indicates:

1. GAS - high food share (lower-income households)
2. TVL - public-sector, younger household heads, lower income
3. STEM - older household heads, higher income, higher education spending
4. ABM - low food share, higher income, private/SUC-LUC sectors

3.3 Summary of Key Insights

1. Socioeconomic Status Strongly Shapes SHS Track Selection

- Higher household income and education spending lead to greater participation in specialized academic tracks (STEM, ABM, A&D).
- Lower-income households and regions with high food share tend to choose GAS or TVL, reflecting financial constraints and accessibility.

2. Regional Clustering Highlights Distinct Household Profiles

- Most regions fall into two main socioeconomic profiles (Cluster 0 vs. Cluster 1).
- Only two outlier regions show exceptionally high academic specialization.

3. Food Share Is the Most Critical Socioeconomic Indicator

- It predicts household capability, educational investment, and SHS track patterns.
- Food share consistently appears in correlation, clustering, and decision tree results.

4. School Sector Significantly Determines Available Pathways

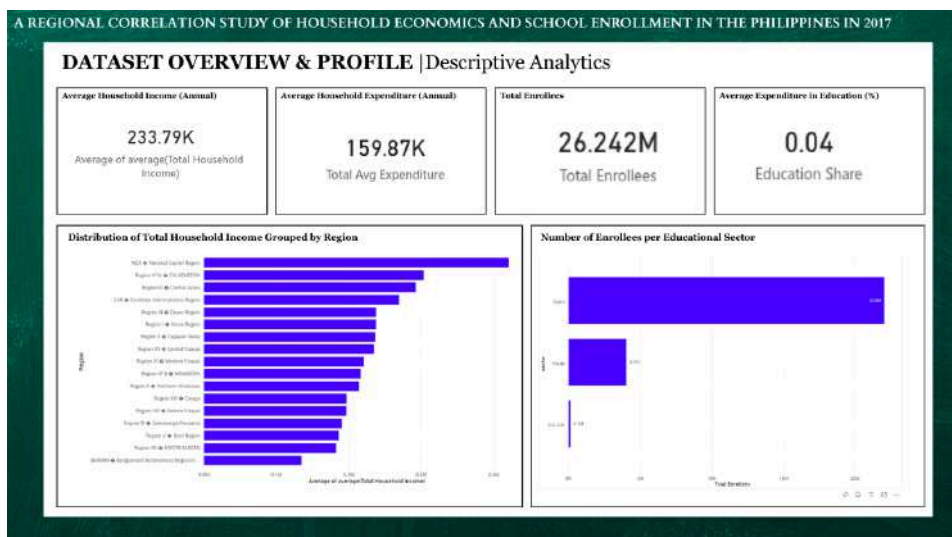
- Public schools tend toward technical-vocational programs (TVL).
- Private and SUC-LUC schools offer broader academic opportunities (STEM, ABM).

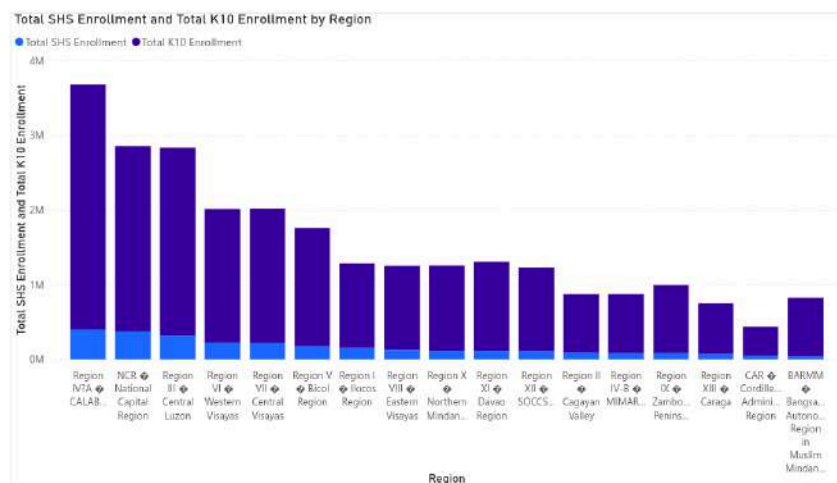
5. Policy-Relevant Insight: Family Investments in Education Shift Track Distribution

- Regions with higher educational budgets per family see greater enrollments in SHS specialization.
- Improving access to academic tracks may require addressing financial and structural constraints, especially in lower-income regions.

IV. Visualizations

4.1 Dashboard



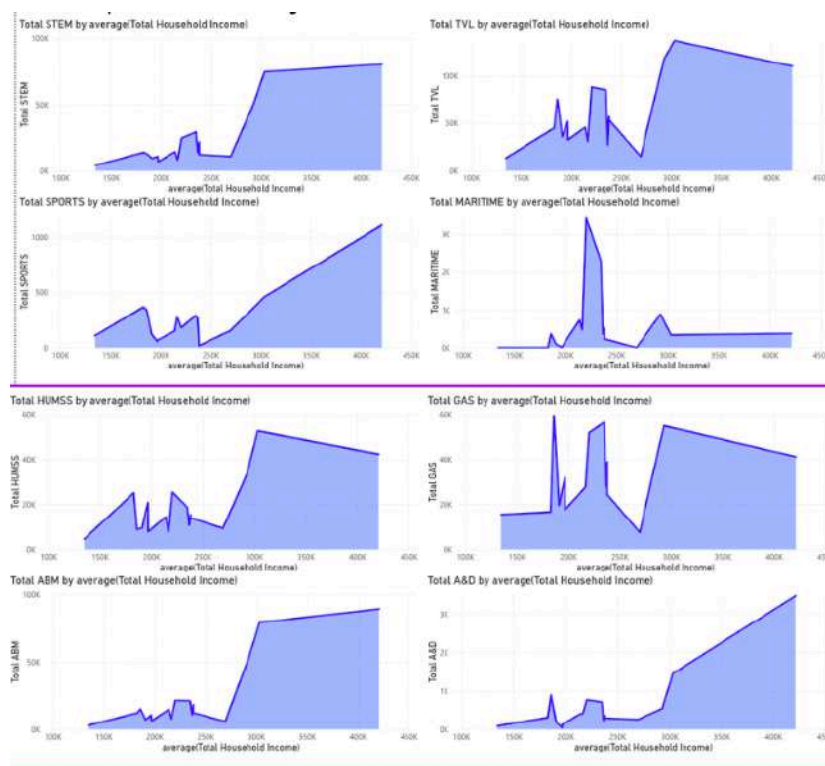




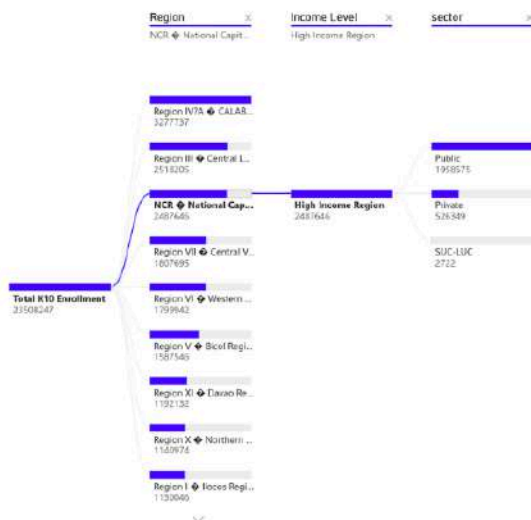
Question 2 & 3. Which socioeconomic factors most strongly influence SHS strand selection? What correlations exist between household socioeconomic variables and school enrollment indicators, such as K–10 totals, SHS totals, and SHS strand counts?

Correlation Matrix (Correlation Matrix [2])																
Attributes	average...	average...	average...	average...	average...	average...	K10_To...	Total_S...	ABM_T...	HUMSS...	STEM...	GAS_T...	TVL_Ta...	MARIT...	SPORT...	A&D_T...
average(Total Household Income)	1	0.991	-0.712	0.712	0.935	-0.499	0.246	0.433	0.540	0.483	0.586	0.121	0.338	0.917	0.642	0.663
average(Education Expenditure)	0.991	1	-0.022	0.622	0.810	-0.370	0.266	0.361	0.464	0.399	0.431	0.060	0.380	-0.036	0.403	0.473
average(Food Share)	-0.712	0.622	1	-1.000	-0.460	0.870	-0.063	-0.161	-0.267	0.219	-0.245	0.067	0.077	0.102	-0.252	-0.256
average(Non-Food Share)	0.712	0.622	-1.000	1	0.460	-0.870	0.063	0.161	0.267	-0.219	0.245	-0.067	0.077	-0.102	0.252	0.256
average(Total Food Expenditure)	0.935	0.810	-0.460	0.460	1	-0.315	0.310	0.595	0.592	0.513	0.563	0.239	0.426	0.943	0.563	0.581
average(Total Number of Family members)	-0.499	-0.370	0.870	-0.670	-0.315	1	-0.081	-0.170	-0.195	-0.198	-0.162	-0.082	-0.159	-0.079	-0.112	-0.129
K10_Total	0.246	0.266	-0.063	0.063	0.310	-0.081	1	0.585	0.235	0.516	0.118	0.776	0.791	-0.128	0.603	0.296
Total_SHS	0.433	0.361	-0.161	0.161	0.596	-0.170	0.585	1	0.887	0.958	0.939	0.888	0.941	0.172	0.761	0.812
ABM_Total	0.540	0.464	-0.267	0.267	0.590	-0.195	0.235	0.887	1	0.896	0.982	0.476	0.091	0.220	0.674	0.871
HUMSS_Total	0.483	0.399	-0.219	0.219	0.513	-0.198	0.516	0.958	0.896	1	0.847	0.895	0.877	0.142	0.681	0.759
STEM_Total	0.586	0.431	-0.245	0.245	0.563	-0.162	0.118	0.939	0.982	0.847	1	0.412	0.010	0.320	0.584	0.832
GAS_Total	0.121	0.060	0.067	-0.067	0.209	-0.082	0.776	0.813	0.476	0.895	0.412	1	0.927	0.803	0.804	0.527
TVL_Total	0.338	0.280	-0.077	0.077	0.420	-0.159	0.791	0.941	0.091	0.877	0.810	0.927	1	0.949	0.741	0.662
MARITIME_Total	0.017	-0.036	0.102	-0.102	0.043	-0.079	-0.128	0.172	0.220	0.142	0.320	0.803	0.949	1	-0.091	0.177
SPORTS_Total	0.642	0.403	-0.252	0.252	0.563	-0.112	0.603	0.761	0.674	0.681	0.584	0.804	0.741	-0.091	1	0.838
A&D_Total	0.663	0.473	-0.256	0.256	0.581	-0.129	0.296	0.812	0.871	0.759	0.832	0.527	0.662	0.177	0.838	1

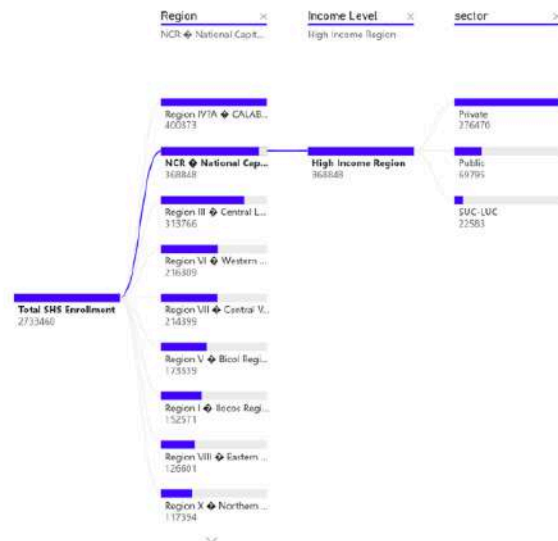
Question 4. Which SHS strands do students from regions with higher household income or higher education expenditure more likely to pursue?



Question 5. & 6. Are there observable associations between income levels and the type of academic sector students enroll in? What insights can be derived from the combined economic–education data that may help policymakers understand regional disparities in schooling and resource needs?

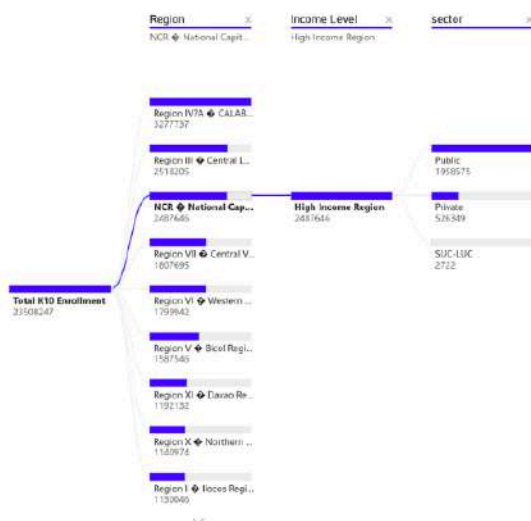


Decomposition of K10 Enrollment Across Regions

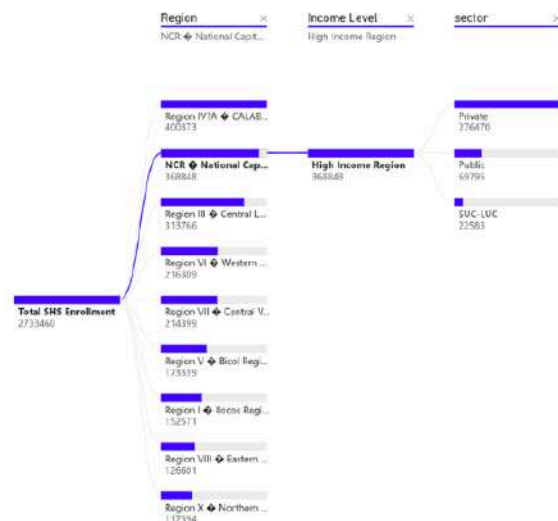


Decomposition of SHS Enrollment Across Regions

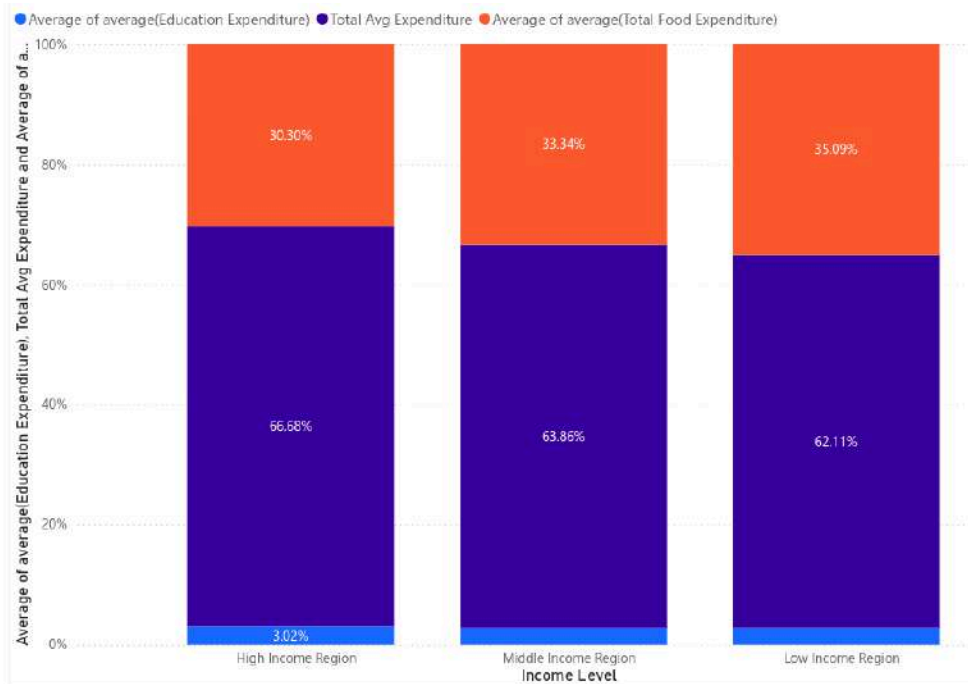
4.3 Insight-Focused Visualization



Decomposition of K10 Enrollment Across Regions



Decomposition of SHS Enrollment Across Regions



Key influencers Top segments

What influences sector to be ?

When...

...the likelihood of sector being Private increases by

Total STEM goes up
12784.56

Total MARITIME goes up
572.15

Total SPORTS goes down
128.77

Total TVL goes down
22479.21

Total HUMSS goes up
7233.00

Infinity

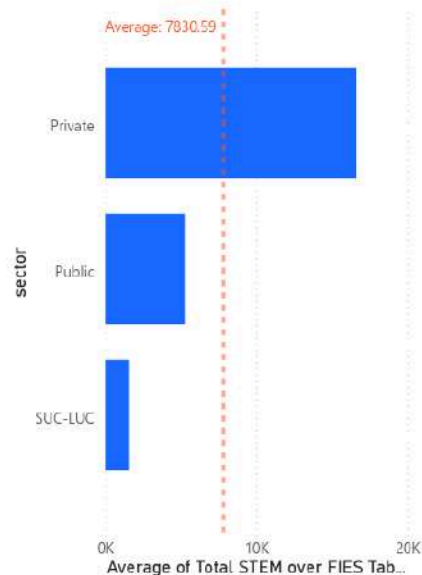
38.66x

28.73x

4.03x

2.20x

On average when Total STEM increases, the likelihood of sector being Private increases.



V. Conclusion and Recommendations

5.1 Summary of Findings

The regional correlation study successfully integrated household socioeconomic characteristics from the Filipino Family Income and Expenditure Survey (FIES) with K–12 school enrollment statistics for Academic Year 2017–2018 at the regional level. The synthesis of descriptive, clustering, and predictive analysis techniques yielded the following key insights:

Descriptive Analytics: Established Core Economic and Enrollment Context

Economic Profile: The average annual household income across all regions was found to be 233.79K PHP, with an average of only 4% of total household expenditure allocated to education, establishing a baseline for financial capacity.

Enrollment Distribution: The majority of the 26.242 million total enrollees were in the Public sector (22.0M), with the National Capital Region (NCR) and Region IV-A CALABARZON identified as having the highest concentrations of both K–10 and SHS enrollment.

Sector Association: Visualization of SHS track enrollment by income revealed that high-income regions, such as NCR, show a disproportionately high reliance on the Private sector for SHS education, contrasting with the overall national public-sector dominance.

Correlation Matrix: Identified Key Socioeconomic Associations

Household Income and Education Spending: Regional average Total Household Income and Education Expenditure are moderately to strongly and positively correlated with enrollment totals in specialized Senior High School (SHS) academic tracks, specifically ABM, STEM, SPORTS, and A&D. This suggests a quantitative link between regional economic capacity and participation in higher-investment educational paths.

Socioeconomic Vulnerability: The derived attribute Food Share was negatively correlated with income-related variables and also negatively correlated with enrollment in the aforementioned specialized tracks, reinforcing that regions allocating a larger portion of the budget to basic sustenance exhibit lower participation in academic SHS tracks.

k-Means Clustering: Profiled Distinct Regional Socioeconomic Types

Cluster 0 (Smaller Household, Lower Vulnerability): This cluster (25 records) was characterized by centroids indicating smaller family sizes and lower food share, suggesting higher socioeconomic stability.

Cluster 1 (Larger Household, Higher Vulnerability): This cluster (21 records) was defined by centroids showing larger family sizes and higher food share, suggesting a lower-income profile distinctly associated with significantly lower enrollment in the ABM strand.

Cluster 2 (High Academic Specialization): This small, outlying cluster (2 records) was defined by centroids with extremely high enrollment counts in A&D, ABM, and STEM, indicating highly urbanized or economically specialized regions.

Decision Tree Classification: Identified Principal Predictors of SHS Track Choice

Food Share as Primary Splitter: The decision tree model established Food Share as the strongest initial predictor of regional SHS track choice. Regions with high food share were most likely assigned to the GAS strand.

Refining Predictors: For regions with low food share, Household Head Age became the next major differentiating factor, with older household heads leading to STEM or ABM, and younger household heads (especially in the Public sector) frequently choosing TVL.

Sector Prevalent Influencers:

Analysis of sector association confirmed that high enrollment in Total STEM increases the likelihood of the sector being Private by 38.66x, while high enrollment in Total TVL decreases the likelihood of the sector being Private, indicating a profound and reciprocal relationship between academic track selection and the type of educational institution available/chosen.

5.2 Recommendations

Based on the data-driven insights that link regional socioeconomic conditions to SHS track selection, the following specific and actionable recommendations are proposed:

Implement Targeted Subsidy Programs for Academic Tracks: Given that high food share and low education expenditure are strong predictors for enrollment in non-specialized tracks (GAS, TVL), policymakers should develop financial assistance specifically for students in Cluster 1 regions (high food share) who wish to pursue high-cost academic tracks (STEM, ABM, A&D). This intervention directly addresses the economic constraint revealed by the decision tree.

Enhance Quality and Employability of Public TVL Programs: Since the Public school sector in lower socioeconomic regions (Cluster 1) shows a distinct inclination toward the TVL strand, resources should be allocated to modernize TVL facilities and curricula within this sector. This includes mandatory industry-relevant certification and job placement assistance to maximize the return on this common educational pathway.

Data-Driven Resource Allocation Model: The defined regional clusters (Cluster 0, Cluster 1, Cluster 2) should be adopted as the basis for the Department of Education's resource distribution. Funding, teacher training, and infrastructure projects should be disproportionately weighted toward the specific needs of Cluster 1 (e.g., K–10 retention, TVL upgrade) and Cluster 2 (e.g., maintaining high academic standards, expanding specialized course capacity).

5.3 Limitations of the Study

The following limitations should be considered when interpreting the results:

Ecological Fallacy (Aggregation Bias): The study relies on regional-level averages for household income and expenditure, and aggregated enrollment counts. This aggregation masks the true household-level relationships, and conclusions drawn at the regional level may not accurately reflect the choices of individual families.

Cross-Sectional Data: The FIES data and the filtered enrollment data are for a single academic year (AY 2017–2018), preventing the establishment of causal relationships or the analysis of time-based trends and policy impacts.

Imputation and Data Cleaning: The decision to replace missing numerical values with 0 or the mean introduces a potential bias, particularly in the household income and expenditure data, which could slightly skew the regional averages.

Missing Variables: The analysis is limited by the variables available in the public datasets. Critical factors such as the distance to school, school quality, cultural values, or specific government social programs are not included, which may have a significant influence on enrollment choices.

VI. Appendices

6.1 Power BI File Link

As the researcher is limited by the web-version of PowerBI due to a difference of operating system. The following is a link to the PowerBI dashboard plain visualizations can be seen in (.pbix): [Google Drive Folder Link](#)

If you wish to see a stylized dashboard, refer to the pdf contained within the folder that used Canva for the design and PowerBI for creating dashboards.

6.2 Raw and Cleaned Data Set Samples

The publicly available datasets used for this project, along with their sources, are provided below for reproducibility:

- Raw Dataset (Filipino Family Income and Expenditure Survey): [Kaggle Dataset Link](#)
- Raw Dataset (Philippines School Enrollment Data): [Kaggle Dataset Link](#)
- Compiled and Cleaned Dataset: [Google Drive Link](#)

6.3 Self Reflection for the Project

Hans Jio Arca

The journey of conducting this data mining project, particularly managing the data preparation and analysis largely on my own, proved to be both incredibly tiresome and surprisingly fun. The initial challenge lay in integrating two distinct, publicly sourced datasets, a task that required careful and methodical cleaning, normalization, and aggregation to ensure that the data granularities were uniform at the regional level. The hours I spent wondering what processes in RapidMiner to use and cross-referencing regional names were taxing, especially since it does not automatically save processes, highlighting the essential, yet often overlooked, complexity of the data preprocessing phase of the CRISP-DM methodology, when it comes to working with tools and such.

However, the reward came during the analysis. Seeing the theoretical models (correlation, clustering, decision trees) produce interpretable insights that directly addressed the study's objectives was a deeply satisfying experience. I was able to answer my study objectives based on my data mining processes. Furthermore, it came as a surprise since I had my initial assumptions, but the data contradicted what I thought was the truth.

Looking back, a significant improvement to the process would have been to shift the initial research focus. Instead of starting from a blank slate, a better approach would be to first conduct an extensive literature review to identify shortcomings or unanswered questions in existing, relevant studies. This proactive research would have provided a stronger theoretical framework and led to a more focused, impactful set of research questions from the outset, potentially guiding the selection of data and techniques more efficiently. Allocating proper, time-boxed phases for each stage of the project and being more open to early-stage experimentation could also enhance the rigor and timeliness of the final report.